AI-Powered Real Estate Property Valuation Assistant

End-to-End System Architecture: Historical Data Processing to Multi-Agent Valuation Response

1. Data Pipeline Layer

- Ames Housing Dataset (OpenML)
- 2,930 Records, 81 Raw Features
- Standardize 15 Key Fields
- Quality & Condition Score Mapping
- Searchable Document Generator
- Export ames_housing_cleaned.csv

2. Hybrid Retrieval Engine

- Dense Vectors: all-MiniLM-L6-v2
- Precomputed Cache (npy format)
- Lexical Search: BM25 / TF-IDF
- Metadata Filtering (Age/Beds/Loc)
- Hybrid Fusion: Score Blending
- Comparable Prioritization Index

3. Predictive ML Service

- Gradient Boosting Regressor
- Random Forest Ensemble
- Benchmark Test $R^2 = 0.9031$
- MAE: ,889 | MAPE: 8.46%
- 90% Prediction Interval Bounds
- Model Persistence (joblib)

4. Risk & Impact Engine

- Physical Age Depreciation Risk
- Quality-Condition Discrepancy
- Neighborhood Volatility Risk
- Layout Outlier Penalty
- Location Premium vs City Median
- What-If Scenario Simulation

5. Autonomous Multi-Agent Collaborative Core (Orchestrator)

- Guardrails Agent: Boundary Checks (100-15k sqft, 1-10 qual) & Adversarial Injection Filter
- Classification Agent: Market Tier (Luxury/Mid/Entry), Typology, Condition Grade, & Demographics
- Valuation Agent: Quality-Adjusted Comps, Comp Median, PPSF Weighting, & Range Estimation
- Risk Agent: Multi-Dimensional 0-100 Risk Score, Vulnerability Flags, & Mitigation Advice
- LLM-as-Judge Agent: Independent Audit, Market Alignment, ML Consistency, & Verdict (APPROVED)
- Synthesis Engine: Reconciled FMV, Confidence Intervals, & Executive Valuation Dossier

6. Microservice API (FastAPI)

- POST /api/v1/search (Hybrid Comps)
- POST /api/v1/valuation/collaborate
- POST /api/v1/valuation/predict-price
- POST /api/v1/valuation/predict-risk
- POST /api/v1/feedback/submit
- GET /api/v1/evaluation/run (DeepEval)

7. Evaluation & Continuous Learning

- DeepEval Quality Benchmark (v4.2.8)
- Metrics: Faithfulness, Answer Relevancy
- Contextual Precision & Recall Tracking
- Appraiser Feedback Loop Repository
- Historical MAE / MAPE Error Tracking
- Dynamic Bias Calibration Offset Factor

8. Presentation Layer (Streamlit)

- Interactive Property Specs Input Form
- Prioritized Comps Table with Deltas
- Reconciled Valuation Dashboard
- What-If Renovation Simulator
- Risk Breakdown & Mitigation Advisories
- Appraiser Feedback Submission Widget